

RESPONSE TO THE R29 ECONOMETRICA REFEREE REPORTS

QIAN QI
Peking University

1. REVIEW OBJECTS AND SUBSTANTIVE REVISION

We address the independent second-pass report at review commit 3175974aed89dac1595af174a201dcae7bcfae31 and the first R29 report at 3f2724d3db622d26f6b1e439fc01311c4858ee7e. Both concern manuscript commit 788246778893695471015ce4db76e6a61a2c9ca0. This is a new R30 scientific object, not a relabeling of R29’s response to R27.

The revision adds regional certification, a minimum-intervention Bellman problem, exact weighted cost enclosures, a frozen new model/cohort, strong structural classical controls, a genuinely active stopping-boundary oracle, and a matched-oracle search comparison. The original whole-domain .01 target and the 47-dimensional derivative/residual bridge are not falsely marked complete. The entire R29 submission object and earlier history are preserved in HISTORY_R30. The item-by-item dispositions below distinguish new results, corrected interpretation, preserved audit material, and uncompleted numerical requirements.

2. INDEPENDENT SECOND-PASS REPORT

2.1. R29-SP-F1: Regional work and strongest classical baseline

Disposition: New conditional complexity theorem; no universal neural speed advantage. Sections 3–6 replace mandatory state enumeration by whole-region inequalities and a geometry-dependent work bound. The operating-mode example certifies the continuum, but its exact symbolic sign solver and online budget guard exploit the same structure. Their inclusion prevents a claim that neural fitting is necessary or generally fastest. The intervention-cost theorem supplies a separate economic objective. The inherited adverse total-cost ratios are retained.

2.2. R29-SP-F2: Universal completion pass rate

Disposition: Corrected interpretation. The abstract and numerical section no longer treat universal post-completion success as approximation efficacy. The raw-policy error bounds, fitting and proof costs, region counts, policy sizes, weighted intervention costs, and strong structural baselines are reported separately. Passing completed objects are implementation consistency checks.

2.3. R29-SP-F3: Raw accuracy at scale

Disposition: New raw evidence; inherited failures unchanged. The new frozen cohort reports raw continuous-domain regret for all 60 candidates, not only for selected seeds. Raw passes and failures are shown by model, width and optimizer. No new result reclassifies the inherited inventory failures. The operating-mode family is a separate one-dimensional structural example, not a rescue of the old high-dimensional accuracy claim.

2.4. R29-SP-F4: Neural-specific value

Disposition: Neural exclusivity not established. The title identifies neural proposals and certified Bellman operators as different objects. Polynomial proposals, the exact symbolic sign solver, the online minimum-intervention guard, Bernstein-based structural solving and vectorized DP receive the same model information. Neural fitting is fully charged. The paper does not claim a method-specific speed advantage contradicted by these controls.

2.5. R29-SP-F5: Economic value and minimum intervention

Disposition: New theorem and exact cost enclosure. Section 4 defines discounted, policy-induced occupancy cost and proves a minimum-cost Bellman recursion within explicit certified action sets. Section 5 bounds the implemented cover cost above and below relative to the local-loss class optimum. Priority weights and exact operating gains replace Hamming distance as the economic comparison. The exact online guard is acknowledged to attain that class optimum; global optimality over every true-regret-feasible policy is not claimed.

2.6. R29-SP-F6: Reference-free terminology

Disposition: Corrected terminology and accounting. The current text uses precomputed-optimum-free and independently implemented exact reference. It expressly charges the complete model, own-policy evaluation, witness construction, range bounds, compilation and checking. Absence of an optimal-value file is not presented as a reduction in model information or Bellman work.

2.7. R29-SP-F7: Scalability and seed coverage

Disposition: New regional-cardinality result, not high-dimensional robustness. The new cover applies to a continuum and hence to midpoint grids with cardinality up to 2^{40} without materializing them. The example is explicitly one-dimensional. It uses five seeds, two widths and two optimizers for each of three gain functions. The inherited dimension-five through dimension-seven runs remain single-configuration pipeline demonstrations; no multidimensional neural robustness theorem is inferred.

2.8. R29-SP-F8: Original full-domain target

Disposition: Original target remains numerically unresolved. The original economy, current-state actor requirement, all-state/all-restart .01 target, and bound 7.181834580823298 remain explicit. The new active-boundary derivative computation is a substantive instantiation where stopping is material, but it does not reduce the inherited global bound or certify the unrestricted actor. The response does not mark this numerical requirement closed.

2.9. R29-SP-F9: Economically material stopping

Disposition: New validated active-boundary calculation. Section 7 moves the restart to preference 1.22 while retaining the original model primitives. The lower-boundary hitting probability at zero adjustment is rigorously enclosed near .6891565168. A killed kernel retains that boundary exactly, and validated Gaussian moments enclose payoff and two derivatives for all five predeclared controls. The failed first nested-integration attempt is retained; it is not promoted to a certificate.

2.10. R29-SP-F10: Polling versus verified optimization

Disposition: New derivative-bracketing comparator; 47-dimensional oracle uninstantiated. The inherited central-state polling schedule is compared with derivative bracketing using the identical validated oracle and concavity audit. Construction, audit, search calls and achieved projected-gradient bounds are separated. Bracketing requires fewer oracle calls in these scalar comparisons. The paper retains polling as a generic certificate wrapper and does not extrapolate it to an instantiated 47-dimensional solver.

2.11. R29-SP-F11: Restricted economic comparative static

Disposition: New boundary curvature result; restricted class stated. The active-boundary calculation yields strictly positive second-derivative intervals near zero adjustment and strictly negative ones at .1 and .2. Thus central-state concavity cannot be transferred to the boundary. Consumption and investment remain fixed in this calculation; it is not an unrestricted comparative static or a proof of interval-wide monotonicity from five points.

2.12. R29-SP-F12: Rollback benefit

Disposition: Preserved in audit material; no net-benefit claim. The complete R29 rollback evidence is reproduced in the historical annex. Its acceptance/restoration contract is retained, but the paired final outcomes are not presented as positive welfare or solver-performance evidence. This revision does not claim a new prospective avoided-loss experiment for the inherited production schedule.

2.13. R29-SP-F13: Historical transport identity

Disposition: Preserved as an implementation identity. The original full-recentering argument and numerical records remain in the historical annex. The main text identifies the same carrier, gradient, Jacobian and Adam history and highlights the roughly 1.39–1.50 omitted nonlinear displacement in the earlier control. Equivalence is not called an alternative optimizer or a computational saving.

2.14. R29-SP-F14: Prospective evidence and dependence

Disposition: Frozen new exploratory cohort; no independence claim. The protocol was committed before the new local scientific runs. All seeds, widths, optimizers and model functions were retained. The amendment records the failed nested integral and stronger post-diagnostic classical controls without changing the candidates. The study is explicitly report-informed and exploratory; neither 60 candidates nor repeated certificates are described as independent economic replications.

2.15. R29-SP-F15: Nearest-neighbor novelty

Disposition: Explicit theorem-level comparison. The introduction compares residual propagation, rollout/policy improvement, approximate policy iteration, baseline-bootstrapped safe improvement and verified numerics. Bellman comparison is identified as classical. The stated mathematical additions are the certified-class cost optimization, whole-region proof obligations and their geometry-dependent work accounting. The strong online guard is included rather than obscured by a novelty claim.

2.16. R29-SP-F16: Coherent paper and preservation

Disposition: Restructured main argument; no historical deletion. The main argument now follows regional certification, economically weighted revision, structural computation and active-boundary diagnosis. The complete prior article, supplement, response, computational appendix and historical annex are reproduced in HISTORY_R30. Original source and evidence directories are not overwritten. Transport and rollback no longer carry independent solver-superiority claims.

2.17. R29-SP-F17: Title and certified object

Disposition: Object-specific title. The title is Certified Bellman Operators with Neural Proposals: Adaptive Verification and Costly Policy Revision. It identifies both the approximation stage and the certified object without attributing a representation-independent guarantee or a repaired policy to a raw neural network.

2.18. R29-SP-T1: Zero-tolerance limit

Disposition: Proposition and exact tests. The zero-tolerance proposition states the finite exact zero-tolerance limit and proves optimality by backward induction. Retained optimal ties need not equal the replacement tie rule. Sixteen small exact policy cases are checked. No claim of zero-tolerance interval-subdivision termination at irrational roots is made.

2.19. R29-SP-T2: Total certified-solution ratios

Disposition: Main-table addition. The article displays both (repair+verification)/DP and (fit+repair+verification)/DP for every inherited row with the necessary recorded times. Missing fitting times are not imputed as zero. Ratios are derived from the displayed R29 inputs and not reported with spurious input precision.

2.20. R29-SP-T3: Edit count versus scan cost

Disposition: New algorithm and explicit distinction. The exhaustive algorithm still has full-scan cost regardless of edits. The new implementation certifies complete intervals, counts every range-oracle call and breakpoint, and separately charges compilation and checking. Its work depends on certificate geometry, not automatically on the number of edited actions.

2.21. R29-SP-T4: Weighted edit metrics

Disposition: Exact occupancy and payoff records. The new records include discounted occupancy edits, state-priority-weighted cost with weight $1+x$, exact raw-to-completed operating-payoff gains, independent overlay checks, and break-even intervention shadow prices. The displayed continuous-density integrals are not mislabeled as finite-grid occupancy estimates.

2.22. R29-SP-T5: Stronger rollout and improvement controls

Disposition: Analytic structural comparison; no new inventory rollout panel. In the new exogenous-transition family every continuation cancels from the action comparison, at every lookahead depth and after any policy-evaluation step. Exact symbolic optimization and the online cost-minimizing guard are therefore stronger controls than a shallow rollout. These are implemented. A new deeper-rollout panel for the inherited inventory family is not claimed.

2.23. R29-SP-T6: Optimized exact baseline

Disposition: Vectorized DP and exact structural controls. The new experiment includes full-grid vectorized backward recursion, explicitly labeled binary64 diagnostic arithmetic, and a genuinely exact symbolic structural policy with zero regret. Both use the same primitives as the neural certificate. The old scalar reference and its timings remain unchanged, and no arithmetic certificate is inferred from floating-point DP output.

2.24. R29-SP-T7: Rational bit complexity

Disposition: Asymptotic bound and observed bit sizes. Section 5 gives a common-denominator bound $Q^{2(T-t)+1}$, bit growth $O(T \log Q + \log((T+1)R))$, and the coarse primitive-denominator version. Rational arithmetic and normalization costs are distinguished from enumeration. Each new cover records its maximum integer bit length and serialized size.

2.25. R29-SP-T8: Principled budget choice

Disposition: Explicit economic design problem. Section 4 defines budget selection over a finite prespecified family: solve minimum revision cost in each nonempty certified class, then select the least-cost class. This is exactly optimal over that union, not an efficient solution to unrestricted budget design. The experiment fixes a common budget prospectively and does not claim a numerically optimized state-budget profile.

2.26. R29-SP-T9: Actual stopping boundary

Disposition: Validated boundary experiment. The same response and evidence as R29-SP-F9 apply: preference 1.22, material lower exit, killed-kernel derivatives, explicit tails and time truncation, and all five predeclared controls. The tiny probability bound pertains only to the opposite boundary.

2.27. R29-SP-T10: 47-dimensional bridge

Disposition: Not falsely closed. The full financed/current-state derivative and residual constants are not supplied by the scalar moment oracle. The original all-domain target and this missing instantiation remain explicit in the article, supplement and machine-readable disposition. No readiness test or passing finite certificate is substituted for the missing mathematical/computational object.

2.28. R29-SP-T11: Hardware deployment

Disposition: Exact compiled object; hardware neural rounding not certified. The compiler and integer-index evaluator use exact rational arithmetic and explicit tie points. Tests link every leaf and endpoint to the frozen mathematical network. The manuscript does not claim that arbitrary hardware floating-point evaluation of the network inherits that guarantee. An additional end-to-end rounding enclosure would be needed for that different deployment object.

2.29. R29-SP-T12: Independent implementation and information

Disposition: Terminology corrected. The reference is described as independently implemented, with the same model information. The new structural controls exploit all the same analytic identities and primitives. Software independence and information independence are not conflated.

3. FIRST R29 REPORT: COMPLETE ITEM CROSSWALK

The following crosswalk addresses every first-report finding, technical comment, presentation comment and numbered next-revision request. Each entry adopts the substantive response and qualification in the identified second-pass item; it does not declare an uncompleted requirement resolved.

3.0.0.1. *R29-F1*. Universal completion pass rate. See R29-SP-F2. Corrected interpretation.

3.0.0.2. *R29-F2*. Regional work and strongest classical baseline. See R29-SP-F1. New conditional complexity theorem; no universal neural speed advantage.

3.0.0.3. *R29-F3*. Raw accuracy at scale. See R29-SP-F3. New raw evidence; inherited failures unchanged.

3.0.0.4. *R29-F4*. Economic value and minimum intervention. See R29-SP-F5. New theorem and exact cost enclosure.

3.0.0.5. *R29-F5*. Neural-specific value. See R29-SP-F4. Neural exclusivity not established.

3.0.0.6. *R29-F6*. Scalability and seed coverage. See R29-SP-F7. New regional-cardinality result, not high-dimensional robustness.

3.0.0.7. *R29-F7*. Original full-domain target. See R29-SP-F8. Original target remains numerically unresolved.

3.0.0.8. *R29-F8*. Polling versus verified optimization. See R29-SP-F10. New derivative-bracketing comparator; 47-dimensional oracle uninstantiated.

3.0.0.9. *R29-F9*. Rollback benefit. See R29-SP-F12. Preserved in audit material; no net-benefit claim.

3.0.0.10. *R29-F10*. Historical transport identity. See R29-SP-F13. Preserved as an implementation identity.

3.0.0.11. *R29-F11*. Nearest-neighbor novelty. See R29-SP-F15. Explicit theorem-level comparison.

3.0.0.12. *R29-F12*. Reference-free terminology. See R29-SP-F6. Corrected terminology and accounting.

3.0.0.13. *R29-F13*. Prospective evidence and dependence. See R29-SP-F14. Frozen new exploratory cohort; no independence claim.

3.0.0.14. *R29-F14*. Scalability and seed coverage. See R29-SP-F7. New regional-cardinality result, not high-dimensional robustness.

3.0.0.15. *R29-F15*. Coherent paper and preservation. See R29-SP-F16. Restructured main argument; no historical deletion.

- 3.0.0.16. *R29-F16*. Title and certified object. See R29-SP-F17. Object-specific title.
- 3.0.0.17. *R29-T1*. Zero-tolerance limit. See R29-SP-T1. Proposition and exact tests.
- 3.0.0.18. *R29-T2*. Total certified-solution ratios. See R29-SP-T2. Main-table addition.
- 3.0.0.19. *R29-T3*. Edit count versus scan cost. See R29-SP-T3. New algorithm and explicit distinction.
- 3.0.0.20. *R29-T4*. Weighted edit metrics. See R29-SP-T4. Exact occupancy and payoff records.
- 3.0.0.21. *R29-T5*. Economic value and minimum intervention. See R29-SP-F5. New theorem and exact cost enclosure.
- 3.0.0.22. *R29-T6*. Stronger rollout and improvement controls. See R29-SP-T5. Analytic structural comparison; no new inventory rollout panel.
- 3.0.0.23. *R29-T7*. Independent implementation and information. See R29-SP-T12. Terminology corrected.
- 3.0.0.24. *R29-T8*. Optimized exact baseline. See R29-SP-T6. Vectorized DP and exact structural controls.
- 3.0.0.25. *R29-T9*. Scalability and seed coverage. See R29-SP-F7. New regional-cardinality result, not high-dimensional robustness.
- 3.0.0.26. *R29-T10*. Rational bit complexity. See R29-SP-T7. Asymptotic bound and observed bit sizes.
- 3.0.0.27. *R29-T11*. Polling versus verified optimization. See R29-SP-F10. New derivative-bracketing comparator; 47-dimensional oracle uninstantiated.
- 3.0.0.28. *R29-T12*. Polling versus verified optimization. See R29-SP-F10. New derivative-bracketing comparator; 47-dimensional oracle uninstantiated.
- 3.0.0.29. *R29-T13*. 47-dimensional bridge. See R29-SP-T10. Not falsely closed.
- 3.0.0.30. *R29-T14*. Restricted economic comparative static. See R29-SP-F11. New boundary curvature result; restricted class stated.
- 3.0.0.31. *R29-T15*. Hardware deployment. See R29-SP-T11. Exact compiled object; hardware neural rounding not certified.
- 3.0.0.32. *R29-T16*. Historical transport identity. See R29-SP-F13. Preserved as an implementation identity.
- 3.0.0.33. *R29-T17*. Historical transport identity. See R29-SP-F13. Preserved as an implementation identity.

3.0.0.34. *R29-T18*. Rollback benefit. See R29-SP-F12. Preserved in audit material; no net-benefit claim.

3.0.0.35. *R29-P1*. Total certified-solution ratios. See R29-SP-T2. Main-table addition.

3.0.0.36. *R29-P2*. Hardware deployment. See R29-SP-T11. Exact compiled object; hardware neural rounding not certified.

3.0.0.37. *R29-P3*. Nearest-neighbor novelty. See R29-SP-F15. Explicit theorem-level comparison.

3.0.0.38. *R29-P4*. Coherent paper and preservation. See R29-SP-F16. Restructured main argument; no historical deletion.

3.0.0.39. *R29-N1*. Coherent paper and preservation. See R29-SP-F16. Restructured main argument; no historical deletion.

3.0.0.40. *R29-N2*. Regional work and strongest classical baseline. See R29-SP-F1. New conditional complexity theorem; no universal neural speed advantage.

3.0.0.41. *R29-N3*. Economic value and minimum intervention. See R29-SP-F5. New theorem and exact cost enclosure.

3.0.0.42. *R29-N4*. Stronger rollout and improvement controls. See R29-SP-T5. Analytic structural comparison; no new inventory rollout panel.

3.0.0.43. *R29-N5*. Neural-specific value. See R29-SP-F4. Neural exclusivity not established.

3.0.0.44. *R29-N6*. Prospective evidence and dependence. See R29-SP-F14. Frozen new exploratory cohort; no independence claim.

3.0.0.45. *R29-N7*. Original full-domain target. See R29-SP-F8. Original target remains numerically unresolved.

3.0.0.46. *R29-N8*. 47-dimensional bridge. See R29-SP-T10. Not falsely closed.

3.0.0.47. *R29-N9*. Rollback benefit. See R29-SP-F12. Preserved in audit material; no net-benefit claim.

3.0.0.48. *R29-N10*. Nearest-neighbor novelty. See R29-SP-F15. Explicit theorem-level comparison.

3.0.0.49. *R29-N11*. Optimized exact baseline. See R29-SP-T6. Vectorized DP and exact structural controls.

3.0.0.50. *R29-N12*. Prospective evidence and dependence. See R29-SP-F14. Frozen new exploratory cohort; no independence claim.